

(Q) Apply Shannon-Fano code & Huffman code for a zero memory source emitting the symbols $\mathcal{S} = \{s_1, s_2, s_3, s_4, s_5\}$ with probabilities

$$P(s_i) = \begin{cases} 2^{-i}, & i = 1, 2, \dots, 4 \\ 2^{-i+1}, & i = 5. \end{cases} \quad K$$

(Soln)

$$P(s_1) = \frac{1}{2}; P(s_2) = \frac{1}{4}; P(s_3) = \frac{1}{8}; P(s_4) = \frac{1}{16}$$

$$P(s_5) = \frac{1}{16}$$

$\mathcal{S}$	$P(\mathcal{S})$				
s_1	$\frac{1}{2}$	0			
s_2	$\frac{1}{4}$	1	0		
s_3	$\frac{1}{8}$	1	1	0	
s_4	$\frac{1}{16}$	1	1	1	0
s_5	$\frac{1}{16}$	1	1	1	1

$$s_1 - 0$$

$$s_2 - 10$$

$$s_3 - 110$$

$$s_4 - 1110$$

$$s_5 - 1111$$

$$L = 1 \times \frac{1}{2} + 2 \times \frac{1}{4} + 3 \times \frac{1}{8} + 4 \times \frac{1}{16} + \frac{4}{16}$$

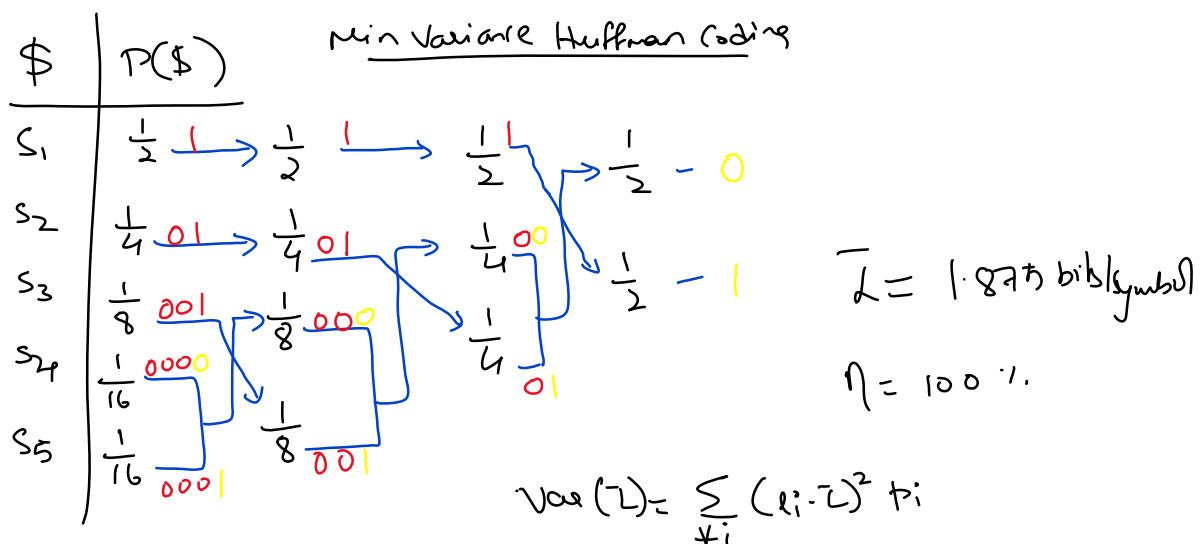
$$= \frac{15}{8} = 1.875 \text{ bits/symbol}$$

$$\log_2 \frac{1}{P(s_i)} =$$

$$H(\mathcal{S}) = - \left[\frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{4} \log_2 \frac{1}{4} + \frac{1}{8} \log_2 \frac{1}{8} + \frac{1}{16} \log_2 \frac{1}{16} + \frac{1}{16} \log_2 \frac{1}{16} \right]$$

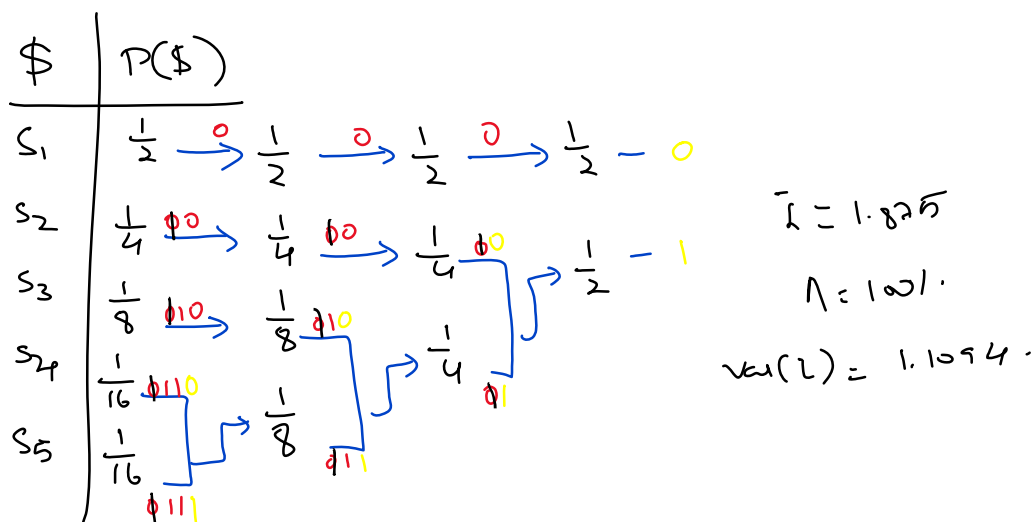
$$= \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \frac{4}{16} = 1.875 \text{ bits/symbol}$$

$$\eta = \frac{H(\mathcal{S})}{L} = 1 \text{ (or) } 100\%$$



$$= (1 - 1.875)^2 \frac{1}{2} + (2 - 1.875)^2 \frac{1}{4} + (3 - 1.875)^2 \frac{1}{8} + 2 \times \frac{1}{16} \times (4 - 1.875)^2$$

$$= 1.1094$$



Dyadic Distribution:

$$\mathcal{S} = \{s_1, s_2, \dots, s_k\}$$

$$P(s_i) = \begin{cases} 2^{-i}, & i = 1, 2, \dots, k-1 \\ 2^{-i+1}, & i = k \end{cases}$$

$\sum_{i=1}^k$

$$\frac{1}{2}, \frac{1}{2^2}, \frac{1}{2^3}, \frac{1}{2^4}, \dots, \frac{1}{2^k}, \frac{1}{2^k}$$

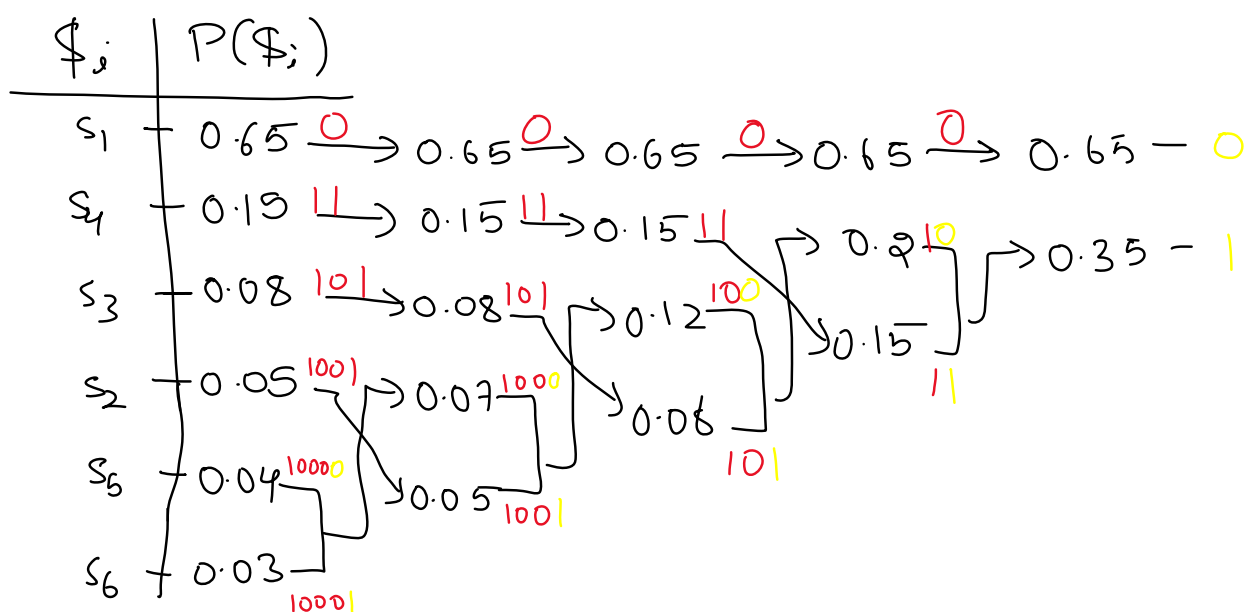
$$\mathcal{S} = \{s_0, s_1, \dots, s_{k-1}\}$$

$$P(s_i) = \begin{cases} 2^{-i-1}, & i = 0, 1, \dots, k-2 \\ 2^{-i}, & i = k-1 \end{cases}$$

(Q) Decode the binary data given by "10001000111001101" into the symbols whose probabilities are shown which are encoded Huffman coding procedure

$\$i$	s_1	s_2	s_3	s_4	s_5	s_6
$P(\$i)$	0.65	0.05	0.08	0.15	0.04	0.03

(Sol:) Perform Huffman Coding



$s_1 - 0$ $s_3 - 101$ $s_5 - 10000$
 $s_2 - 1001$ $s_4 - 11$ $s_6 - 10001$

The given binary data is "10001000111001101"

$\underline{s_6}$ $\underline{s_1 s_1 s_1 s_4}$ $\underline{s_2}$ $\underline{s_3}$

The decoded data is " $s_6 s_1 s_1 s_1 s_4 s_2 s_3$ ".

Code Type:

